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Data repository:https://github.com/AimBtlv/research-NetworkCitation_Edduba.git **(./ 8-output_Modul1/)**

Bibliographic Networks of Knowledge Transmission in Edubba Scholarship

Abstract

This project applies directed citation network analysis to a corpus of 2,103 academic publications on Edubba scribal schools of the Old Babylonian period. A corpus was constructed from bibliographic citation expansion and modelled as a directed scholarly network.A custom Step-Distance Model[1] helps us to classify data and stratifies each node by its structural proximity to empirical sources, enabling a hypothesis-driven investigation of five interconnected structural properties. Using centrality measures, custom step-distance classification, and trust-triangle detection, the study identifies how primary archaeological data, mediating interpretations, and authoritative scholarship are structurally related within academic discourse. Results reveal that the citation network is not hierarchical but radically sparse and fragmented; that primary sources are structurally isolated; that a bottleneck of 37 mediators (1.8%) controls all information pathways; that epistemic influence concentrates in Step 3 authorities rather than primary data; and that structurally reliable sources are extremely rare (less 1%) and belong to the mediator layer, not the authority layer. These findings demonstrate that network analysis can operationalise epistemic concepts such as reliability and mediation in a quantitatively tractable and reproducible form.

Introduction

The study of Edubba schools, the scribal institutions of the Old Babylonian period (20th–17th centuries BCE) presents a characteristic methodological problem in historical disciplines: a large body of secondary scholarship has accumulated around a limited and fragmentary empirical base of excavated educational tablets. While thousands of cuneiform tablets have been archaeologically documented, only a fraction of the literature can be traced back to direct archaeological evidence. The remaining majority reproduces, reinterprets, or synthesises established narratives whose empirical origins are difficult to recover through linear reading alone. This study does not aim to contribute directly to Assyriology. Instead, it treats the scholarly corpus as a data object: a directed citation network in which structural analysis can reveal how knowledge is organised, transmitted, and stabilised. The research question is not what is known about Edubba, but how that knowledge circulates and where its structural foundations lie. This approach is grounded in the tradition of computational social science[2] and

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[1] More detailed description can be found in the github repository (./7-documentation/) **docStep-Distance Model.md**

[2] Lazer et al., 2009

bibliometric network analysis, in which citation networks are modelled as graphs G = (V, E), where V represents publications and E directed citation edges. Within this framework, standard graph-theoretic measures degree distribution, betweenness centrality, eigenvector centrality, and community structure provide a quantitative basis for investigating epistemic organisation. A central contribution of this project is the Step-Distance Model: a custom classification scheme that assigns each node an epistemic role based on its structural position relative to primary archaeological data. This model operationalises the concept of epistemic reliability as a network property, enabling systematic hypothesis testing.

Research Questions and Hypotheses

RQ1 - How is scholarly knowledge about Edubba schools structurally organised within the academic citation network and what role do primary archaeological sources play within this structure?

RQ2 - Which sources can be identified as epistemically reliable based on their structural position relative to primary data and do the most reliable sources coincide with the most cited ones?

To address these questions, five interconnected hypotheses were formulated. Each hypothesis tests a specific structural claim and builds on the result of the preceding one, forming a sequential chain of evidence.

H1. Network Topology Hypothesis -The citation network has a hierarchical (tree-like) structure in which primary sources form the root layer, mediators the intermediate layer, and authorities the terminal layer.

H2.Structural Gap Hypothesis - Primary archaeological sources(Step 1) are structurally isolated from the most cited authoritative works (Step 3), there are no direct or indirect citational links between them.

H3.Bottleneck Hypothesis - The only structural bridge between Step 1 and Step 3 is an extremely small layer of mediators (Step 2), whose role constitutes a network bottleneck that is a structural property, not an artefact of the classification method.

H4.Epistemic Displacement Hypothesis - Structural influence (eigenvector centrality) is concentrated in Step 3 authorities, not in Step 1 primary source--s. Signific-ance is formed where interpretations are repeatedly cited, not where data originates.

H5.Epistemic Reliability Rarity Hypothesis - Sources that simultaneously satisfy all three reliability criteria connection to primary data, high in-degree, and mutual reinforcement via a Trust Triangle are structurally rare (less 1% of the corpus) and belong to the mediator layer (Step 2) rather than the authority layer (Step 3).

Data

Corpus Construction

A seed corpus of approximately 150 anchor sources was compiled on the basis of two selection criteria: first exclusive thematic scope - publications dedicated to scribal education in the Old Babylonian period and second unique DOI identifiers, ensuring verifiability and reproducibility. Large heterogeneous monographs were excluded, exceptions were made for works exclusively focused on Mesopotamian scribal practices. In parallel, archaeological excavation reports were collected as Step 1 primary sources and stored in a structured CSV file (sourcesArchaeologicalRelies.csv) containing site metadata, excavation details, tablet counts, and bibliographic references. These were entered manually, as DOI coverage for excavation reports is incomplete

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Bibliographic Expansion — Snowball Sampling

The anchor corpus was expanded using a snowball bibliographic method. PDF documents were converted to text using pdfplumber (digital files) and pytesseract OCR (scanned materials). Bibliographies were automatically extracted using AnyStyle, converted to structured JSON, and matched against the existing corpus to identify citing–cited relationships. The resulting dataset contains 2,103 nodes and 2,524 directed edges.

Data Abstraction[3]

The dataset is modelled as a directed graph G = (V, E) where each node v ∈ V represents an academic publication and each directed edge (u to v) ∈ E represents a citation from u to v. This relational abstraction shifts the analytical frame from the attributes of individual publications to the topology of their connections. The primary analytical interest lies in three derived properties: degree distribution (structural prominence), betweenness centrality (mediation), and eigenvector centrality (embeddedness in densely interconnected clusters).

Analytical Abstraction Layer

Step-Distance Model

To move beyond citation counts and assess epistemic structure, each node was assigned a step_distance value (1–5) based on a combination of network-theoretic criteria. This model operationalises epistemic proximity to primary data as a structural graph property.

Distance	Source Role	Source Function	Citation Network Method	Edge Quantity	Percentage distribution
Step_1	Direct sources	Publications providing direct empirical evidence — excavation reports, tablet publications, and site documentation	Filled Manually	164	7.8%
Step 2	Mediators	Interpretive works that cite primary data and transmit empirical findings into the broader scholarly discourse structural bridges between evidence and interpretation"	betweenness ∈ top 15% **AND** cites ≥ 1 Step1 nodes	37	1.8%
Step 3	Authorities (High in-degree)	Canonical sources within this corpus works whose interpretations have stabilised and are repeatedly cited as authoritative reference points	in-degree ∈ top 10% **AND** cited by ≥ 2 Step2 nodes **AND** publication age ≥ threshold (10 years)	226	10.7%

[3] More detailed description can be found in the github repository (./7-documentation/) docDataAbstraction.md

Step 4	Mediated source	Synthetic works - comparative studies and reviews that consolidate authority-level interpretations without direct grounding in primary evidence	cites no Step1 **AND** cites ≥ 2 Step3 **AND** out-degree ≥ 2	3	0.1%
Step_5	Peripheral	All others	**NOT** direct reliance **NOT** Mediators **NOT** Authority **NOT** Mediated Sources	1673	79.6%

Table 1. Step-Distance Model: epistemic role assignment criteria and corpus distribution (full graph, N = 2,103).

Methods

Analytical Pipeline

The analytical workflow follows a sequential hypothesis-driven design, structured as a pipeline of five stages: (1) graph construction, (2) centrality computation, (3) subgraph extraction, (4) community detection, and (5) hypothesis testing. Each stage produces outputs that serve as inputs for the next, forming a reproducible analytical chain.

Graph Construction

A directed citation graph was constructed using igraph (Python). The full corpus yields a graph with 2,050 nodes and 2,524 directed edges, with a global density of 0.0006 and reciprocity ≈ 0.0008 indicating a nearly acyclic structure consistent with one-directional citation flow. The graph decomposes into two weakly connected components, confirming structural fragmentation at the macro level. Due to extreme sparsity, a subgraph was extracted for detailed analysis by filtering nodes with betweenness centrality in the top 15% (N = 400, 841 edges, density = 0.0053). This subgraph preserves the structurally active portion of the network while removing peripheral nodes that contribute no relational signal.

Centrality Measures

In-degree centrality counts the number of incoming citation edges for each node. In a citation network, high in-degree indicates that a source has been adopted as a reference point by many other works a proxy for canonisation[4].

Betweenness centrality measures the fraction of shortest paths between all pairs of nodes that pass through a given node. High betweenness identifies structural brokers: nodes whose removal would disconnect or restructure large portions of the graph[5]. In the epistemic context of this study, high betweenness corresponds to interpretive mediation, the transmission of empirical findings through the citation chain.

Eigenvector centrality assigns each node a score proportional to the sum of the scores of its neighbours. Unlike in-degree, eigenvector centrality rewards embeddedness in densely interconnected, mutually reinforcing clusters. In a knowledge network, it approximates "discursive authority" the influence of a work within the epistemic core[6].

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[4] Easley & Kleinberg, 2010

[5] Freeman, 1977

[6] Bonacich, 1987

PageRank is a variant of eigenvector centrality that accounts for the direction and weight of edges, originally designed for web link analysis[7]. It was computed as a supplementary measure to cross-validate eigenvector findings.

Community Detection

Community structure was detected using the Louvain algorithm[8], which optimises modularity to identify groups of nodes with higher internal connectivity than expected under a random graph model. The detection was applied to the epistemic subgraph (N = 400) to identify epistemic clusters and examine whether community membership correlates with step-distance roles.

Sensitivity Analysis

To assess whether the bottleneck structure identified in H3 is a genuine structural property rather than an artefact of the betweenness filter used to define Step 2, a sensitivity analysis was performed. The betweenness criterion was relaxed and Step 2 membership was recomputed. This test verified that the set of nodes citing Step 1 sources remains stable regardless of the filter threshold.

Results

H1. Network Topology: The Graph Is Not a Hierarchy

Analysis of the step-distance distribution immediately contradicts the hierarchical model. Step 5 (peripheral nodes) accounts for 79.6% of the corpus (n = 1,673), while Steps 2 and 4 together contain only 40 nodes (less 2%). If the network were hierarchical, the distribution across Steps 1–4 would be roughly balanced. Instead, the near-total dominance of the peripheral layer indicates a radically flat, sparse structure. Global graph metrics confirm this: density = 0.0006, reciprocity = 0.0008, two weakly connected components. The graph is not a tree embedded in a knowledge hierarchy but a sparse, fragmented network with several dense local clusters and a large disconnected periphery. H1 is not confirmed.

[7] Page et al., 1999
[8] Blondel et al., 2008

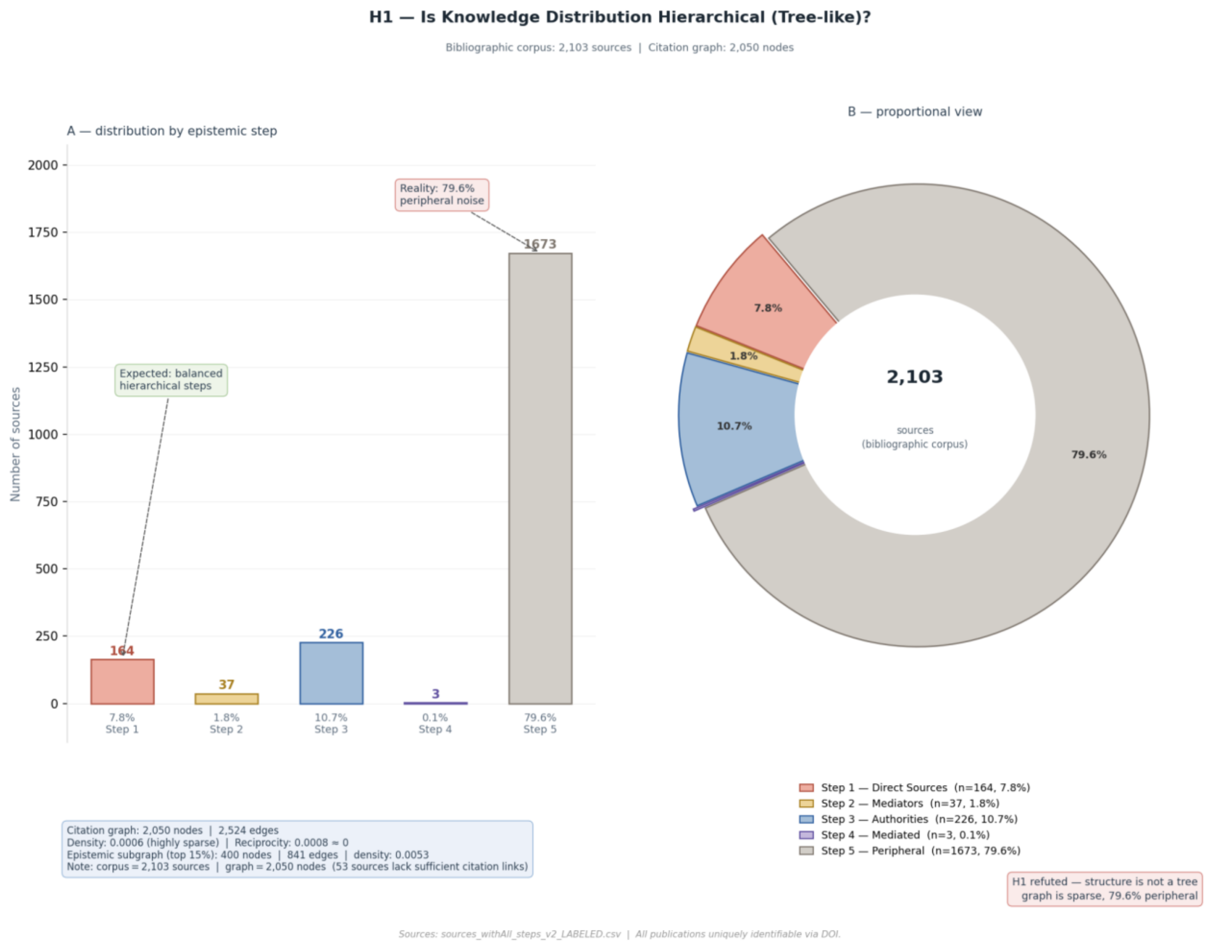

Figure 1. Global Graph: Step Distance Distribution (full graph, N = 2,103).

This result necessitates extraction of the epistemic subgraph (N = 400, density = 0.0053) for subsequent analysis, as the full graph does not provide sufficient relational density for meaningful centrality comparison.

H2. Structural Gap: Primary Sources Are Isolated

Betweenness centrality was computed for all nodes in the epistemic subgraph (N = 400). The results show a stark stratification: 100% of Step 1 nodes (n = 137) have betweenness = 0.00, meaning no primary source lies on any shortest path between any other pair of nodes in the network. Primary sources are structural dead ends: they receive citations (mean in-degree = 1.33, max = 9 for Gasche1989) but transmit no information further through the network. Step 3 authorities are similarly absent from the flow: 99.6% have betweenness = 0. The mean betweenness of Step 2 (94.43) contrasts sharply with Step 1 (0.00) and Step 3 (0.17). There are no direct edges from any Step 1 node to any Step 3 node. H2 is confirmed.

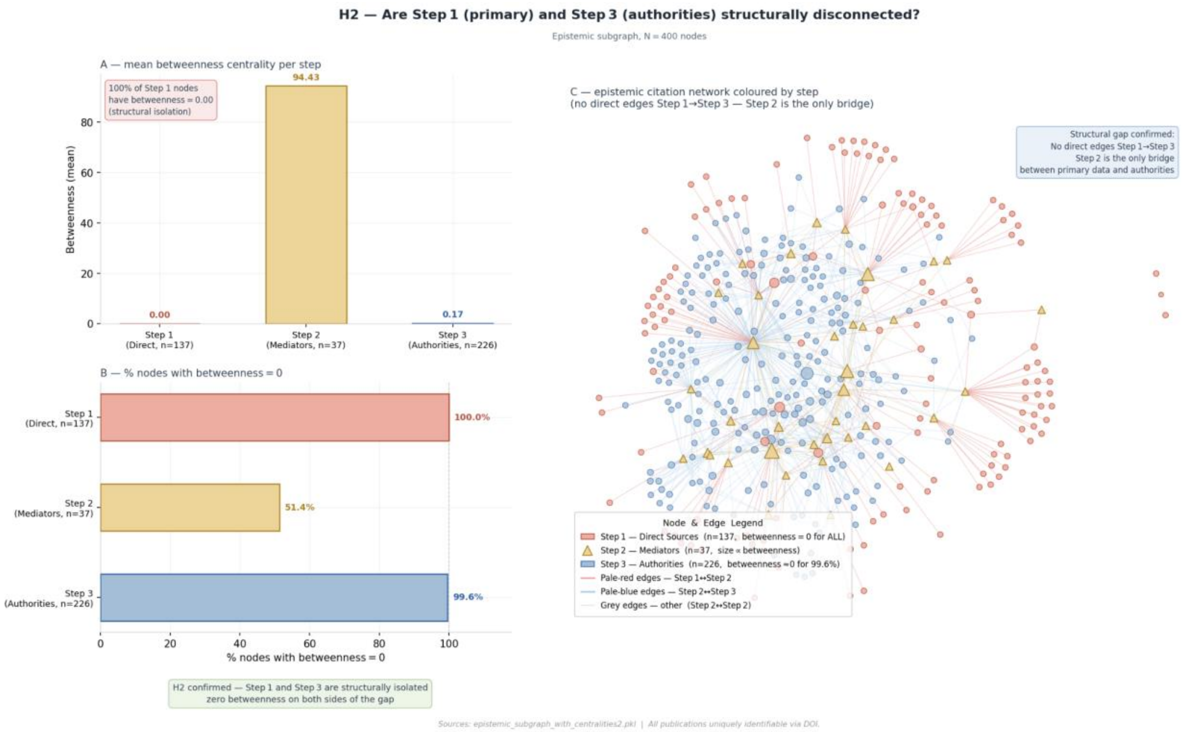

Figure 2. Betweenness Centrality

H3. Bottleneck: 37 Mediators Control All Paths

H2 establishes structural isolation but not the absence of any path between Step 1 and Step 3. H3 tests whether a mediating layer bridges the gap, and whether this structure is robust. The top-5 nodes by betweenness are all Step 2 (Robson2001: 802.17, Cammarosano2014: 551.17, Pearce1995: 541.25, Charpin2010: 421.50, George0000: 376.25), and all top-10 betweenness nodes belong to Step 2.

```
--------------------------------------------------------------------------------
Step 2 (n=37) - single layer with high betweenness
--------------------------------------------------------------------------------
Robson2001         Betweenness 802.17,   In-degree 9
Cammarosano2014    Betweenness 551.17,   In-degree 3
Pearce1995         Betweenness 541.25,   In-degree 3
Charpin2010        Betweenness 421.50,   In-degree 2
George0000         Betweenness 376.25,   In-degree 14
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Sensitivity analysis:
with betweenness filter = 37 nodes; without filter = 37 nodes (unchanged)
----------------------------------------------------------------------------------------------------
Step1 loaded: 137 nodes
Computing centralities...
Step2 (mediators): 37
Step2 (mediators, no_betweenness filter): 37
Step3 (authorities): 226
Step4 (mediated): 3
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Sensitivity analysis confirms robustness: with the betweenness filter applied, Step 2 contains 37 nodes; with the filter removed, Step 2 contains 37 nodes identical. The bottleneck is not a consequence of the classification threshold but a structural property of the graph. Only 37 nodes (1.8% of the corpus) serve as the exclusive transmission channel between primary data and canonical interpretation. **H3 is confirmed.**

H4.Epistemic Displacement Hypothesis

Eigenvector centrality analysis reveals that structural influence is decoupled from empirical proximity. Step 3 authorities show the highest mean eigenvector (0.103), while Step 1 primary sources show a moderate mean (0.059), and Step 2 mediators the lowest (0.035). Critically, 80.3% of Step 1 nodes have eigenvector = 0, meaning they are not embedded in any densely interconnected cluster. This result is consistent with the network model: eigenvector centrality rewards nodes connected to other highly connected nodes. Since Step 1 nodes have low in-degree and are structurally isolated, they cannot accumulate eigenvector mass regardless of their empirical importance.

One anomaly was identified: Kramer (1949) is a Step 1 node with eigenvector = 1.0 (maximum in the subgraph), with in-degree = 7. "Schooldays" is among the earliest descriptive texts on Edubba and has been adopted as a reference by highly central Step 3 nodes in community 5. Crucially, however, its betweenness = 0.0 it is a cited dead end, not a transmission bridge. **H4 is confirmed with the noted anomaly.**

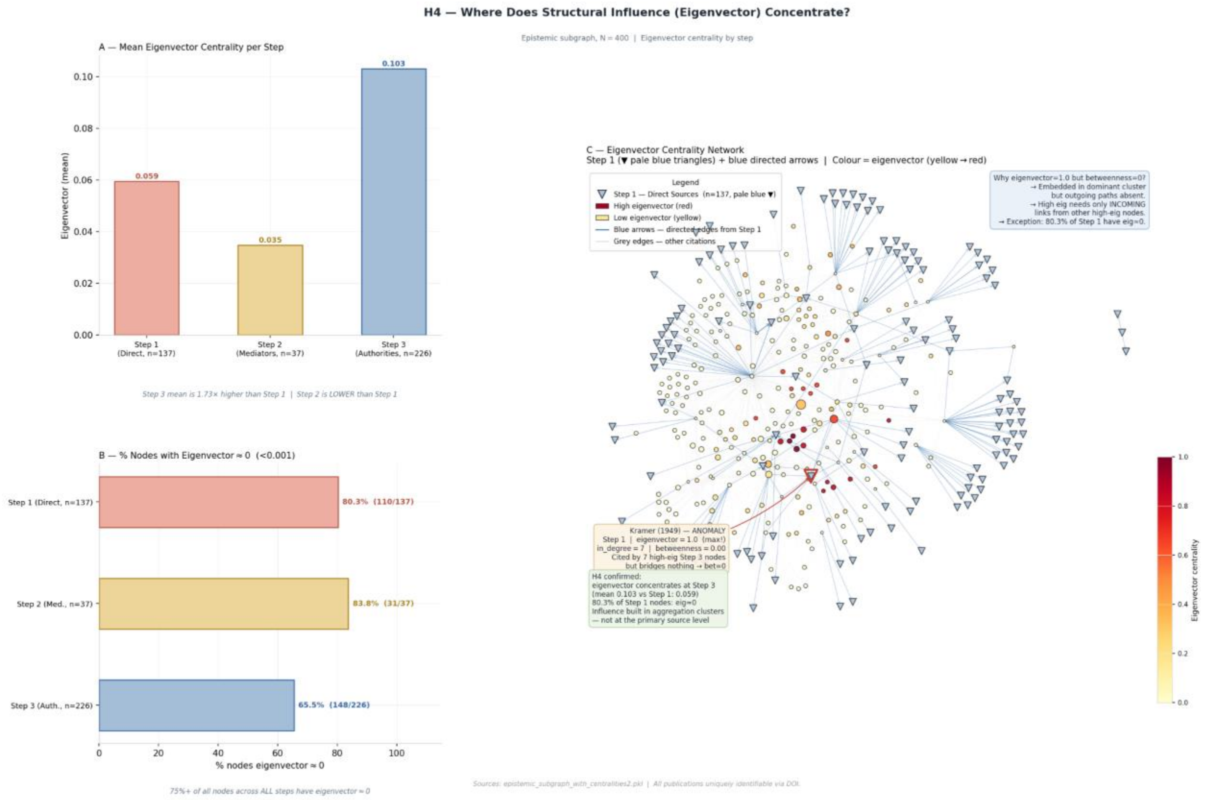

Figure 3. Eigenvector Centrality

Community Structure

Louvain community detection on the epistemic subgraph identified 203 communities. The distribution is extremely skewed: one community (community 5) contains 184 nodes, while the remaining 202 communities contain between 1 and 3 nodes each. This pattern, a single giant component surrounded by many micro-clusters, is consistent with the scale-free properties described in Barabási and Albert (1999) and indicates extreme fragmentation in the scholarly discourse on Edubba. The dominant community (184 nodes) concentrates the majority of Step 3 authorities and the anomalous Step 1 node (Kramer 1949), suggesting that canonical Edubba scholarship has coalesced into a single internally dense epistemic cluster, while the broader field remains fragmented.

H5. Epistemic Reliability Rarity: Reliable Sources Are Structural Mediators

The Trust Triangle criterion defines reliability as a composite structural property requiring three simultaneous conditions: (1) direct citation of at least one Step 1 source, (2) high in-degree, (3) existence of a second node with high in-degree that cites the same Step 1 source, forming a closed triad. This definition operationalises reliability not as a scalar attribute but as a topological configuration. Applying this criterion to the full dataset (N = 2,103) yields two strictly reliable sources (less 0.1% of the corpus): Robson (2001) and George (year unknown). A relaxed ranking produces 10 candidate sources, all belonging to Step 2. No Step 3 authority satisfies the full Trust Triangle, because the definition of Step 3 (in-degree ≥ top 10%) de facto excludes direct citation of Step 1 nodes: the most cited work, Adams (1969, in-degree = 21), has betweenness = 0 and no direct link to primary data.

H5 is confirmed.

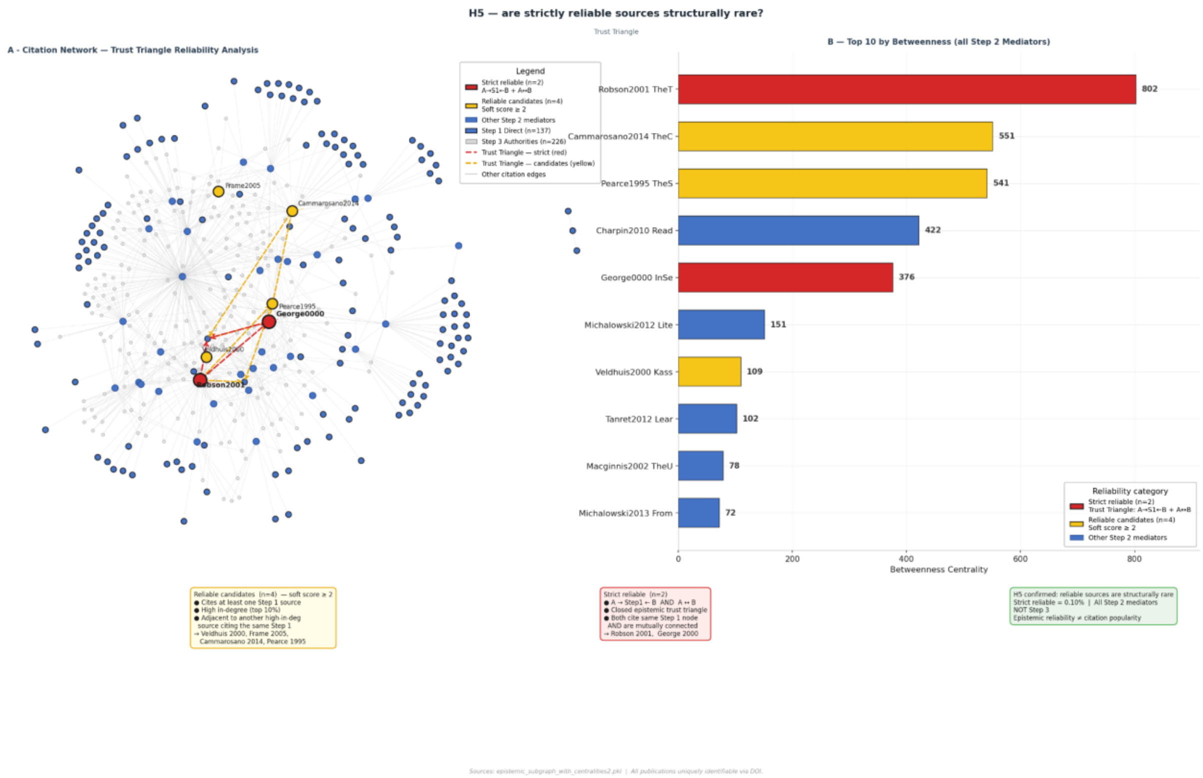

Figure 4. Trust Triangle -Reliable Sources.

Citation popularity and epistemic reliability are structurally distinct properties of the network. The most cited works are not the most reliable, and the most reliable works belong to a narrow, structurally specific layer that is the least visible to traditional bibliometric analysis.

Discussion

The five hypotheses, taken together, reveal a consistent structural pattern: the citation network of Edubba scholarship is not a knowledge hierarchy but a sparse, mediation-dominated graph in which epistemic reliability and citation authority are decoupled.

The bottleneck finding (H3) has implications beyond the specific domain. It suggests that in specialised humanities subfields, the entire chain of knowledge transmission from primary data to canonical interpretation may depend on an extremely small set of intermediary publications. The removal or misrepresentation of these nodes - Robson2001, George0000, Cammarosano2014 would structurally sever the connection between empirical evidence and established scholarly consensus.

The eigenvector displacement finding (H4) resonates with broader debates in bibliometrics and science studies about the relationship between citation frequency and epistemic quality. The network evidence supports the view that discursive influence in academic fields is shaped more by integration into densely connected interpretive clusters than by proximity to empirical data.

The community structure a single dominant cluster of 184 nodes surrounded by 202 micro-communities suggests that canonical Edubba scholarship has undergone a consolidation process consistent with the formation of an epistemic community in the sense described by Haas[9] . Whether this consolidation reflects genuine empirical consensus or narrative lock-in driven by citation momentum cannot be determined from network structure alone, but the question is empirically testable through content analysis.

The Trust Triangle operationalises reliability as a closed relational configuration a triadic closure which has formal precedents in social network theory[10] . Applying this concept to knowledge networks treats the mutual reinforcement of authoritative claims anchored to the same primary source as a structural analogue to the social trust generated by shared acquaintances.

Limitations and Biases

The step-distance classification assigns epistemic roles based on citation structure rather than content analysis. A source is classified as a mediator because it structurally bridges Step 1 and the broader network, not because its claims have been independently verified. The model therefore measures structural position, not epistemic validity.

Step 4 (mediated sources) is severely underrepresented (n = 3), most likely due to the DOI-based sampling strategy, which favours journal articles over textbooks and encyclopedic

[9] Haas (1992)

[10] Simmel, 1908; Granovetter, 1973

works. Synthetic works that consolidate canonical interpretations without citing primary data directly are likely undersampled.

Step 5 absorbs both genuinely peripheral publications and artefacts of corpus incompleteness works that would acquire structural roles if the snowball expansion had been extended. The 79.6% peripheral rate should therefore be read as an upper bound on peripherality rather than a definitive measurement.

The analysis is based on citation structure as extracted from bibliographies, it does not account for negative citations, self-citations, or the semantic content of citations. Future work could extend this framework through natural language processing to distinguish confirmatory from contradictory citation relationships.

Conclusion

This project demonstrates that directed citation network analysis can reveal structural properties of scholarly knowledge that are not accessible through traditional bibliometric or close-reading approaches. By modelling a corpus of 2,103 publications on Edubba scribal schools as a directed graph and applying standard network-theoretic measures, the study identifies five structural findings of methodological significance.

The network is not hierarchical but sparse and fragmented, with a single dominant epistemic community and 202 micro-clusters. Primary archaeological sources are structurally isolated from the canonical literature, connected only through a bottleneck of 37 mediators. Epistemic influence, as measured by eigenvector centrality, is concentrated in Step 3 authorities rather than primary data. And reliable sources defined by the composite Trust Triangle criterion are structurally rare and belong to the mediator layer, not the authority layer.

These findings challenge the assumption that citation frequency is a valid proxy for epistemic reliability. They also demonstrate the value of custom classification models (Step-Distance) that translate domain-specific epistemic concepts into quantitatively tractable network properties. The pipeline developed in this project from data collection and bibliographic expansion through graph construction, centrality computation, community detection, and hypothesis testing is fully reproducible and applicable to other specialised humanities corpora.

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Software and Digital Tools

- **Python** — data processing and analysis (https://www.python.org)
- **igraph** — network construction and analysis (https://igraph.org)
- **AnyStyle** — bibliographic reference extraction (https://anystyle.io)
- **pdfplumber** — PDF text extraction (https://github.com/jsvine/pdfplumber)
- **pytesseract** — OCR for scanned documents(https://github.com/madmaze/pytesseract)
- **hslpicker** - color analyses and design (https://hslpicker.com/?utm_source=chatgpt.com#efb1a6)

Вкладка 2

Что реально интересного обнаружено в данных

1. Надёжные источники — это медиаторы, не авторитеты: Все 4 источника с `reliable_source=1` — Step 2:

- Robson (2001), George (год неизвестен), Veldhuis (2000), Frame (2005) Это **противоречит интуитивному ожиданию** — самые надёжные структурно не самые цитируемые.

2. Аномалия Step 1 в eigenvector: Kramer (1949) — Step 1, но eigenvector = **1.0** (максимум!). Это исключение полностью опровергает H1 в сильной формулировке и требует отдельного объяснения в репорте.

3. Betweenness — почти исключительно Step 2: Топ-10 по betweenness — 9 из 10 узлов это Step 2. Robson2001 — betweenness=802, George0000 — 376. Это подтверждает H4 количественно и очень убедительно.

4. Сообщества — почти все изолированные: 203 сообщества, из которых **одно (community=5) содержит 184 узла**, а остальные 202 сообщества содержат 1–3 узла. Это говорит об экстремальной фрагментации сети — важная находка.

5. Расхождение Step1: 164 vs. 137 — 27 первичных источников есть в полном датасете, но не вошли в подграф. Это стоит упомянуть как limitation.

6. 96% источников — "не primary text" поле `is_primary_text` практически не заполнено. Это проблема данных, которую надо честно ук